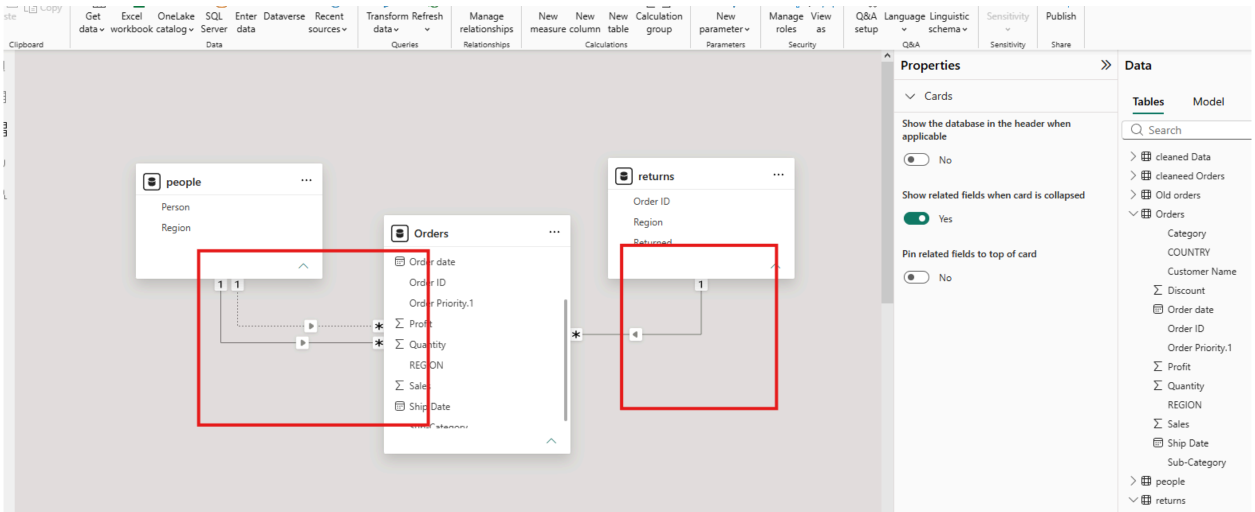
TASK 3 – DATA MODELING

- Identify Fact table(s) and Dimension table(s).
- Build relationships (one-to-many / many-to-one).
- Check cardinality and cross-filter direction.
- ✅ Task: Create a clear and correct data model.



Summary

- Identified **Orders** as the **Fact Table** because it contains transactional data such as Sales, Profit, Quantity, and Order ID.
- Identified **People** and **Returns** as **Dimension Tables** because they store descriptive information (Region, Person, and Return Status).
- Created a **One-to-Many (1:*)** relationship between **People[Region]** and **Orders[REGION]**.
- Created a **One-to-Many (1:*)** relationship between **Returns[Order ID]** and **Orders[Order ID]**.
- Used **Single Cross Filter Direction** to ensure efficient filtering and improve model performance.

Observations

- The **Orders** table acts as the central table, connected to all dimension tables.
- The **People** table enables analysis of sales by region and sales representative.
- The **Returns** table helps identify returned orders and analyze their impact on sales and profit.
- A **star-schema-like model** makes reports easier to build and improves query performance.
- Maintaining a single active relationship between related tables avoids ambiguity and ensures accurate filter propagation.